

CSA0501
DATABASE MANAGEMENT SYSTEMS
LAB EXPERIMENT

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EXP 8:

AIM

To perform subquery and correlated query on the given relations.

ALGORITHM

1. Start.
2. Create a database and select it.
3. Create the STUD2 and FACULTY tables with appropriate columns.
4. Insert sample records into both tables.
5. Write a subquery to compare each student's marks with the average marks (using SELECT AVG(...) as an inner query).
6. Write a query to filter students who wrote more than one assessment.
7. Write a query using ORDER BY and LIMIT to find the most recently joined faculty.
8. Write a subquery combined with ORDER BY to list course-wise marks greater than the average.
9. Display and verify the output of each query.
10. Stop.

COMMANDS (short format)

sql

```
CREATE DATABASE dbms_lab8;
```

```
USE dbms_lab8;
```

```
CREATE TABLE stud2 (...);
```

```
CREATE TABLE faculty (...);
```

```
INSERT INTO stud2 VALUES (...);
```

```
INSERT INTO faculty VALUES (...);
```

```
SELECT * FROM stud2 WHERE marks > (SELECT AVG(marks) FROM stud2);
```

```
SELECT name FROM stud2 WHERE assements > 1;
```

```
SELECT * FROM faculty ORDER BY Doj DESC LIMIT 1;
```

```
SELECT courseno, marks FROM stud2 WHERE marks > (SELECT AVG(marks) FROM stud2) ORDER BY courseno;
```

QUESTIONS

Sub-Query and Correlated Sub-Query

1. Which of the student's score is greater than the average score?
2. Which of the students' have written more than one assessment test?
3. Which faculty has joined recently and when?
4. List the course and score of assessments that have the value more than the average score each course.

OUTPUTS:

```
mysql> SELECT * FROM stud2
-> WHERE marks > (SELECT AVG(marks) FROM stud2);
+-----+-----+-----+-----+-----+-----+
| regno | name  | courseno | marks | facultydoj | assements |
+-----+-----+-----+-----+-----+-----+
| 2001  | Kiran | C01      | 78    | 2010      | 3         |
| 2003  | Vikram | C02      | 82    | 2015      | 4         |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.04 sec)
```

```
mysql> SELECT name FROM stud2
-> WHERE assements > 1;
+-----+
| name  |
+-----+
| Kiran |
| Sneha |
| Vikram |
+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM faculty
      -> ORDER BY Doj DESC
      -> LIMIT 1;
```

FacNo	FacultyName	gender	Dob	Doj	Mobileno	DeptNo
F03	Karthik	M	1990-10-03	2022-01-10	9988771234	IT

```
1 row in set (0.02 sec)
```

```
mysql> SELECT courseno, marks FROM stud2
      -> WHERE marks > (SELECT AVG(marks) FROM stud2)
      -> ORDER BY courseno;
```

courseno	marks
C01	78
C02	82

```
2 rows in set (0.00 sec)
```

RESULT

The records from the tables are displayed using Sub-Query and Correlated Sub-Query.